

Final evidence matrix favors a cautious null-leaning frame

Evidence stream	Manuscript use	Support status	Caution status
D1 original/resting aperiodic candidate	background/rationale only, not a confirmatory result	weak initial signal	exact original rows unavailable; artifact-controlled rerun attenuated
D1 artifact-trimmed acute mTBI	primary cautionary result	no	yes
D1 strict artifact-clean branch	artifact-control limitation and sensitivity note	no	yes
D3 eyes-closed alpha/IAF	negative sensitivity endpoint	no	yes
chronic TBI exploratory branch	separate exploratory subsection	weak context only	batch caveat prevents proof
ds003490 pipeline rehearsal	methods/QC comparator only if needed	no	not applicable
D2 ds005114 DPX cue-baseline	bounded falsification result, not validation	weak partial	insufficient alone
D2 ds005114 task-average	negative/generalization check	no	yes
D2 ds003523 VWM task-average	negative/generalization check	no	yes
D2 mixed-effects models	bounded falsification result	no	yes
D2 within-subject stability/direction consistency	methods/reliability context	measurement only	does not support disease specificity

Green/blue cells mark weak or measurement-only support; red cells mark cautionary or weakening evidence. Source rows are in the paired figure source-data CSV.